

BETH ISRAEL DEACONESS MEDICAL CENTER

Division of Cardiovascular Medicine — Electrophysiology Laboratory

330 Brookline Avenue, Boston, MA 02215

Phone: (617) 667-2920 | Fax: (617) 667-2945

Medtronic CareLink™ Network Remote Monitoring

Cardiac Device Interrogation Report

Patient Name:	CB70E6AE [REDACTED]
Date of Birth:	October 8, 2086
Sex:	Male
Medical Record Number:	MRN-2086-CB70E6AE
Interrogation Date:	July 2, 2150
Interrogation Time:	10:42 AM
Interrogation Type:	In-Office Follow-Up
Performed By:	Jennifer L. Martinson, RN, CCDS
Reviewed By:	David R. Holbrook, MD, FHRS

Device Information

Device Type:	Dual-Chamber Pacemaker
Manufacturer:	Medtronic, Inc.
Model:	Azure XT DR W2DR01
Serial Number:	PZW3045827
Implant Date:	March 14, 2144
Implant Indication:	Symptomatic bradycardia with sinus node dysfunction; paroxysmal atrial fibrillation
Implanting Physician:	Alan M. Kessler, MD, FACC

Battery Status

Battery Voltage:	2.78 V
Battery Status:	Normal Operation (EOS not reached)
Estimated Longevity:	4.2 years (based on current usage)
Elective Replacement Indicator (ERI):	Not reached
Magnet Response:	Enabled (DOO mode at 85 bpm)

Lead Measurements

Right Atrial Lead

Lead Model:	Medtronic CapSureFix Novus 5076-58
-------------	------------------------------------

Serial Number: PJN3124758
Impedance: 448 Ω (Normal range: 200–1500 Ω)
Capture Threshold: 0.75 V @ 0.4 ms
P-Wave Amplitude: 3.2 mV
Pacing Configuration: Bipolar
Sensing Configuration: Bipolar

Right Ventricular Lead

Lead Model: Medtronic CapSureFix Novus 5076-52
Serial Number: PJN3124802
Impedance: 612 Ω (Normal range: 200–1500 Ω)
Capture Threshold: 0.5 V @ 0.4 ms
R-Wave Amplitude: 12.8 mV
Pacing Configuration: Bipolar
Sensing Configuration: Bipolar

Programmed Parameters

Parameter	Current Setting
Mode	DDDR
Base Rate	60 bpm
Maximum Tracking Rate	130 bpm
Maximum Sensor Rate	120 bpm
Paced AV Delay	200 ms
Sensed AV Delay	170 ms
Atrial Amplitude	3.0 V @ 0.4 ms
Atrial Sensitivity	0.5 mV
Ventricular Amplitude	2.5 V @ 0.4 ms
Ventricular Sensitivity	2.8 mV
PVARP	250 ms
Rate Response	ON (Moderate)
Mode Switch	ON
Managed Ventricular Pacing (MVP)	ON

Pacing Statistics (Since Last Interrogation: April 10, 2150)

Total Hours Monitored: 2,014 hours
Atrial Pacing (Ap): 18% (362 hours)
Ventricular Pacing (Vp): 4% (81 hours)
Atrial Sensing (As): 73%
Ventricular Sensing (Vs): 92%
MVP Success Rate: 96%
Average Ventricular Rate: 68 bpm
Maximum Sensor Rate Achieved: 114 bpm (during exertion on 06/18/2150)

Arrhythmia Summary

Atrial High-Rate Episodes (Mode Switch Episodes)

Total Episodes: 14 episodes
Total Duration: 38 hours 22 minutes
Longest Episode: 6 hours 42 minutes (May 22, 2150, 2:14 AM – 8:56 AM)
Average Episode Duration: 2 hours 44 minutes
Burden (% of monitored time): 1.9%

Representative Episode Details:

Date	Start Time	Duration	Max Atrial Rate
May 22, 2150	2:14 AM	6 hr 42 min	152 bpm
June 3, 2150	11:38 PM	4 hr 18 min	148 bpm
June 17, 2150	6:22 AM	2 hr 55 min	144 bpm
June 28, 2150	9:10 PM	5 hr 12 min	158 bpm

Ventricular High-Rate Episodes

Episodes Detected: None

Patient-Triggered Events

Patient Activations: 0

Diagnostics Review

Heart Rate Histogram

The patient's intrinsic heart rate is predominantly in the 60–80 bpm range during the review period. There is minimal nocturnal bradycardia requiring pacing. Daytime heart rates appropriately increase during activity with sensor-driven rate response. No evidence of inappropriate tachycardia.

Lead Integrity

No lead integrity alerts. No evidence of lead fracture, insulation breach, or dislodgement. Impedance trends stable over the past six months. Capture thresholds remain low with adequate safety margin.

Remote Monitoring Compliance

Scheduled Transmissions: Weekly
Transmissions Received: 12 of 12 (100%)
Last Transmission: June 30, 2150

Clinical Interpretation and Recommendations

OVERALL DEVICE PERFORMANCE: Normal. The Medtronic Azure XT DR dual-chamber pacemaker is functioning appropriately with stable lead performance and adequate battery longevity.

BATTERY STATUS: Normal operation. Estimated 4.2 years of longevity remain based on current pacing burden. Elective replacement indicator (ERI) has not been reached. Next in-office evaluation recommended in 12 months or sooner if indicated by remote monitoring alerts.

LEAD PERFORMANCE: Excellent. Both right atrial and right ventricular leads demonstrate stable impedances within normal limits and low capture thresholds with good sensing. No evidence of lead malfunction. Current programmed output settings provide adequate safety margins (4:1 for atrial lead, 5:1 for ventricular lead).

PACING BURDEN: Low. Atrial pacing at 18% and ventricular pacing at 4% reflect appropriate intrinsic conduction with pacemaker backup as needed. The Managed Ventricular Pacing (MVP) algorithm is functioning well, minimizing unnecessary right ventricular pacing. No changes recommended at this time.

ATRIAL ARRHYTHMIAS: Documented paroxysmal atrial fibrillation. Fourteen mode-switch episodes were recorded over the past 84 days, with a cumulative burden of 1.9% and a maximum episode duration of 6 hours 42 minutes. This is consistent with the patient's known history of paroxysmal atrial fibrillation. Episodes are predominantly nocturnal. The patient is currently on anticoagulation (documented long-term warfarin or direct oral anticoagulant use per problem list). Rhythm control with current medical management appears acceptable. Patient should be counseled regarding symptom awareness and timely reporting of palpitations, dizziness, or dyspnea.

VENTRICULAR ARRHYTHMIAS: None detected during the review period. No episodes of non-sustained or sustained ventricular tachycardia.

PROGRAMMING CHANGES: None made at this interrogation. Current DDDR programming with rate-responsive pacing and mode-switch functionality is appropriate for the patient's clinical status and activity level.

RECOMMENDATIONS:

- Continue current pacemaker settings.
- Continue remote monitoring via CareLink with weekly scheduled transmissions.

- Patient to remain on anticoagulation for stroke prevention in atrial fibrillation (CHA₂DS₂-VASc score applicable).
- Reinforce importance of medication compliance, particularly anticoagulation, rate control agents, and antiarrhythmic therapy as prescribed.
- Next in-office device check scheduled for January 2151 (6 months) or sooner if remote monitoring alerts are triggered.
- Patient counseled on symptoms of atrial fibrillation, importance of reporting prolonged palpitations, and signs of stroke.
- Electrophysiology follow-up appointment scheduled for September 2150 with Dr. Holbrook to reassess atrial fibrillation burden and consider adjustments to antiarrhythmic therapy if indicated.

Electronically Reviewed and Approved By:

David R. Holbrook, MD, FHRS

Clinical Cardiac Electrophysiologist

Division of Cardiovascular Medicine

Beth Israel Deaconess Medical Center

Date: July 2, 2150

Time: 3:18 PM

This report was generated from interrogation data obtained via the Medtronic CareLink™ Network. The information contained herein reflects device function at the time of interrogation. This document is intended for the exclusive use of the treating physician and authorized healthcare personnel.